

Uniform Smoothing and Commensurate Observation Masks

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Abstract

Universal comparison of commensurate tilted-uniform observation masks of unequal cardinalities reduces to positivity of an explicit signed cardinal spline. We prove that a finite signed lattice measure becomes nonnegative after sufficiently many convolutions with a unit uniform precisely when its generating polynomial is positive on the positive real axis. The sufficiency is a direct consequence of classical Pólya positivity and the half-unit digit decomposition of a uniform variable. Consequently, adding sufficiently many base-cap observations to the source makes universal comparison possible exactly when the associated denominator polynomial divides the numerator. The criterion is finite and arithmetic. For the quotient $1 - u + u^2$, we give an exact Bernstein certificate showing that the least smoothing order is nine, whereas every order at most eight fails.

1 Unequal numbers of commensurate observations

Under a reference law Q , let the independent coordinates have densities

$$q_{\beta,c}(x) = \frac{e^{-\beta x}}{Z_\beta(c)} \mathbf{1}_{(0,c)}(x), \quad Z_\beta(c) = \int_0^c e^{-\beta u} du, \quad (1)$$

with a common real tilt β . The total of all coordinate caps is finite. An arbitrary independent real noise may be added to the total T ; no moment condition is imposed on it. A mask E universally dominates a mask L when one Markov kernel from the E -observation to the L -observation works under every probability law

$$\frac{dP_w}{dQ} = \frac{w(T)}{\mathbb{E}_Q w(T)}, \quad w \geq 0, \quad 0 < \mathbb{E}_Q w(T) < \infty.$$

The kernel observes the source mask alone. Masks may overlap. One kernel for every positive-probability lower-threshold conditioning is equivalent to this full weight formulation [4].

Let μ_E, μ_L be the reference laws of the two observed sums. We use the mask convolution criterion

$$E \longrightarrow L \iff \mu_E = \mu_L * \nu \text{ for a probability measure } \nu. \quad (2)$$

This includes overlapping masks and independent noise [4]. To recall why, the conditional vector law given its sum is unchanged by total-sum weighting. The sum experiments therefore suffice. A universal sum kernel, together with complement independence and cancellation of the noise characteristic function near zero, gives

$$\mathbb{E}[e^{zX} f(Y)] = \frac{F_E(z)}{F_L(z)} \mathbb{E}[e^{zY} f(Y)]$$

near zero, where X is the source sum and Y is the kernel output with law μ_L . Taking $f(y) = e^{-zy} g(y)$ shows that $X - Y$ is independent of Y , first near zero in transform space

and then everywhere by compactness. This gives (2). Conversely a factorization permits the conditional kernel of the target summand given its sum with the independent factor. Complement transforms cancel to give the matching target experiment. The factor is unique and compactly supported.

Suppose the finite source mask has m caps $ha_1, \dots, ha_m$ and the target has n caps $hb_1, \dots, hb_n$, where $h > 0$ and the cap integers are positive. Empty lists are allowed. Define

$$\begin{aligned} r &= m - n, & [a]_u &= 1 + u + \dots + u^{a-1}, & \rho &= e^{-\beta h}, \\ A(u) &= \prod_{i=1}^m [a_i]_u, & C(u) &= \prod_{i=1}^n [b_i]_u, & P(u) &= A(u)/C(u). \end{aligned} \quad (3)$$

Write $\mu_{\beta, h}$ for the base-cap law in (1).

Theorem 1.1 (Exact criterion at a fixed cardinality). *Universal dominance holds precisely when $r \geq 0$, C divides A as polynomials, and the following signed measure is nonnegative:*

$$\nu_\beta = \mu_{\beta, h}^{*r} * \left(\frac{1}{P(\rho)} \sum_j p_j \rho^j \delta_{hj} \right), \quad P(u) = \sum_j p_j u^j. \quad (4)$$

The candidate has total mass one and is the unique possible factor. At $r = 0$ its positivity is exactly coefficientwise nonnegativity of P .

Proof. The block decomposition of a cap into intervals of length h gives

$$F_{\beta, ha}(z) = F_{\beta, h}(z) \frac{[a]_{e^{h(z-\beta)}}}{[a]_{e^{-\beta h}}},$$

and hence near zero

$$\frac{F_E(z)}{F_L(z)} = F_{\beta, h}(z)^r \frac{P(e^{h(z-\beta)})}{P(\rho)}. \quad (5)$$

At $z = \beta + 2\pi i/h$, each coordinate transform has exactly one simple zero. The entire-factor identity $F_E = F_L F_\nu$ forces $m \geq n$.

Every root ω of C is a root of unity different from one. Choose z_0 with $e^{h(z_0-\beta)} = \omega$. The base transform $F_{\beta, h}$ does not vanish there, and the exponential map has nonzero derivative. Comparing zero multiplicities forces A to vanish at ω at least as often as C . Thus C divides A . Monic division gives $P \in \mathbb{Z}[u]$, with constant coefficient and leading coefficient both one.

Now (5) is exactly the transform of (4), whose mass is one and whose denominator is positive because $P(\rho) = A(\rho)/C(\rho) > 0$. Compact-transform uniqueness proves necessity of positivity. If it is positive, the same transform identity proves the convolution factorization, and (2) gives sufficiency. $\square$

2 The signed spline and a finite positivity test

Let λ be uniform probability measure on $[0, 1]$, let B_r be the density of λ^{*r} for $r \geq 1$, and put $\lambda^{*0} = \delta_0$. For $P(u) = \sum_j p_j u^j$, set

$$\eta_P = \sum_j p_j \delta_j, \quad f_{r, P}(x) = \sum_j p_j B_r(x - j) \quad (r \geq 1). \quad (6)$$

For $r \geq 1$, the candidate in (4) has density

$$\frac{h^{r-1} e^{-\beta x}}{Z_\beta(h)^r P(\rho)} f_{r, P}(x/h). \quad (7)$$

Indeed each of the r uniform densities contributes its exponential tilt, and the tilt in each lattice coefficient supplies the missing shift factor. This proves that the sign is independent of β . Nonnegativity is understood almost everywhere; for $r \geq 2$ the spline is continuous and the condition is equivalent to pointwise nonnegativity.

The cardinal-spline formula is

$$B_r(x) = \frac{1}{(r-1)!} \sum_{k=0}^r (-1)^k \binom{r}{k} (x-k)_+^{r-1}. \quad (8)$$

At $r = 1$, interpret y_+^0 as $\mathbf{1}_{\{y>0\}}$; knot values do not affect the law. Formula (8) follows inductively by convolution over $[0, 1]$, since integrating a truncated power gives the difference of the next two shifted powers and the binomial recurrence combines their coefficients.

Let

$$c_k = [u^k](1-u)^r P(u).$$

On a unit cell $x = j + t$, $0 < t < 1$,

$$(r-1)! f_{r,P}(j+t) = \sum_{k=0}^j c_k (j+t-k)^{r-1}. \quad (9)$$

The degree- $(r-1)$ Bernstein coefficients of this polynomial are

$$b_{j,\ell} = \sum_{k=0}^j c_k (j-k)^{r-1-\ell} (j+1-k)^\ell, \quad 0 \leq \ell \leq r-1, \quad (10)$$

with $0^0 = 1$. This is the binomial expansion of

$$(v+t)^{r-1} = [v(1-t) + (v+1)t]^{r-1}$$

in the nonnegative basis $\binom{r-1}{\ell} t^\ell (1-t)^{r-1-\ell}$. Consequently nonnegative coefficients in (10) give an exact positivity certificate. If they have mixed signs, formula (9) still permits a complete test by polynomial sign determination on $[0, 1]$. For rational data this is decidable using exact real-root isolation or Sturm sequences. Only the finitely many cells in the support $[0, r + \deg P]$ are needed.

3 Eventual positivity from the theorem of Pólya

The next result is a direct consequence of classical Pólya positivity [1, 2]. The half-unit decomposition makes its continuous-uniform application elementary.

Theorem 3.1 (Uniform smoothing criterion). *For a nonzero real polynomial P , the following are equivalent:*

1. $P(u) > 0$ for every real $u > 0$;
2. $\lambda^{*r} * \eta_P$ is a nonzero nonnegative measure for some integer $r \geq 0$;
3. the same is true for every sufficiently large integer r .

Proof. For real t , the bilateral Laplace transform of a nonzero nonnegative compact measure is strictly positive. The transform in condition 2 is $F_\lambda(t)^r P(e^t)$, and $F_\lambda(t) > 0$. Thus condition 2 implies condition 1.

Conversely assume condition 1. Remove an initial monomial factor u^k if necessary; it only translates the signed measure. We may therefore suppose $p_0 > 0$. Positive-axis positivity also forces the leading coefficient to be positive. A positive constant polynomial is immediate, so let $d = \deg P \geq 1$. The homogeneous form

$$H(s, t) = \sum_{j=0}^d p_j s^{2j} t^{2d-2j}$$

is strictly positive on $s, t \geq 0, s + t = 1$. In the interior this is $t^{2d} P((s/t)^2) > 0$; at the two endpoints it is a positive extreme coefficient. Pólya's theorem supplies N for which

$$(1 + u)^N P(u^2) \tag{11}$$

has nonnegative coefficients, and in fact every coefficient can be made positive.

Let $\lambda_{1/2}$ be uniform probability measure on $[0, 1/2]$ and $\epsilon = (\delta_0 + \delta_{1/2})/2$. The digit decomposition is

$$\lambda = \lambda_{1/2} * \epsilon.$$

Therefore

$$\lambda^{*N} * \eta_P = \lambda_{1/2}^{*N} * \left(2^{-N} \sum_{\ell} [u^\ell] ((1 + u)^N P(u^2)) \delta_{\ell/2} \right). \tag{12}$$

Both measures on the right are nonnegative. Convolving with further copies of λ proves condition 3. Its implication to condition 2 is immediate. $\square$

A self-contained coefficient justification. For the precise Pólya step above, put $D = 2d$ and $n = N + D$. The coefficient of u^k in (11), divided by $\binom{n}{k}$, equals

$$\sum_{j=0}^d p_j \frac{(k)_{2j} (n - k)_{D-2j}}{(n)_D}, \tag{13}$$

where $(v)_s = v(v-1)\cdots(v-s+1)$. This follows by dividing $\binom{N}{k-2j}$ by $\binom{n}{k}$, with out-of-range binomial coefficients zero. Uniformly for $0 \leq k \leq n$, expression (13) converges to $H(k/n, 1-k/n)$: divide each of its finitely many factors by n , whose errors are uniformly $O(1/n)$, and note that $(n)_D/n^D \rightarrow 1$. The limiting form has a positive minimum on the closed simplex, so every coefficient is eventually positive. This is the classical uniform-coefficient mechanism of Pólya's theorem, included to make the required step explicit.

4 Stabilization by adding base-cap observations

Adding k independent base-cap h coordinates to the source observation changes the cardinality difference from r to $r + k$. It leaves $P = A/C$ unchanged because $[1]_u = 1$. The same argument applies when such independent coordinates are already available but previously unobserved.

Corollary 4.1 (Exact stabilization criterion). *For finite commensurate masks, some finite number of additional base-cap source observations makes universal dominance possible if and only if C divides A . Equivalently,*

$$\#\{i : d \mid a_i\} - \#\{j : d \mid b_j\} \geq 0 \quad \text{for every integer } d \geq 2. \tag{14}$$

It suffices to test d up to the largest cap integer.

Proof. The polynomial divisibility obstruction in Theorem 1.1 is unchanged by additional base-cap coordinates. Conversely, a polynomial quotient $P = A/C$ is strictly positive on the positive real axis, since both products of q -integers are positive there. Theorem 3.1 gives a sufficiently large total smoothing order. The equivalence with (14) follows from the classical factorization

$$[a]_u = \prod_{\substack{d|a \\ d>1}} \Phi_d(u)$$

into cyclotomic polynomials. $\square$

A terminating exact algorithm. First test (14); if it fails, no finite addition suffices. Otherwise compute P , and search $N \geq 0$ until every coefficient of $(1+u)^N P(u^2)$ is nonnegative. Theorem 3.1 proves termination and certifies every total smoothing order at least N .

To find the minimum number of added observations, test the admissible total orders

$$s = \max(0, r), \dots, \max(N, r). \quad (15)$$

For $s = 0$ inspect the coefficients of P ; for $s \geq 1$ use the finitely many rational cell polynomials in (9). Exact real-root isolation determines their signs between their roots, including endpoints by continuity when needed. If s_* is the first successful order, the answer is $k_* = s_* - r$. The lower limit in (15) enforces $k \geq 0$, including when the original source already has more coordinates than the target. Once an order succeeds, every later order succeeds by convolution.

5 An exact threshold of nine uniform convolutions

Take source caps h times the list consisting of one 6 and $r + 1$ copies of 1, and target caps $(2h, 3h)$. The cardinality difference is r , and

$$P(u) = \frac{[6]_u}{[2]_u[3]_u} = 1 - u + u^2.$$

At $\beta = 0$ and $h = 1$, the signed candidate measure is

$$\lambda^{*r} * (\delta_0 - \delta_1 + \delta_2),$$

with density $f_r(x) = B_r(x) - B_r(x-1) + B_r(x-2)$ for $r \geq 1$.

Theorem 5.1 (Sharp smoothing order). *The candidate is nonnegative if and only if $r \geq 9$. Consequently, for every real common tilt, precisely nine additional base-cap observations are necessary and sufficient to make the source $(h, 6h)$ universally dominate the target $(2h, 3h)$.*

Proof. The order-eight density satisfies

$$\begin{aligned} 7!f_8(5) &= 5^7 - 9 \cdot 4^7 + 37 \cdot 3^7 - 92 \cdot 2^7 + 154 = -34, \\ f_8(5) &= -\frac{17}{2520} < 0. \end{aligned}$$

It is continuous, so the corresponding measure is not nonnegative. If any order $0 \leq r \leq 8$ were nonnegative, convolving it with $\lambda^{*(8-r)}$ would make order eight nonnegative. Thus all those orders fail.

For order nine, the coefficients of $(1-u)^9(1-u+u^2)$ are

$$(1, -10, 46, -129, 246, -336, 336, -246, 129, -46, 10, -1).$$

Formula (10) gives the exact certificate in Table 1. All coefficients are nonnegative. Since $B_9(x) = B_9(9 - x)$, the density f_9 is symmetric about $11/2$, so the first six cells cover the remaining cells by reflection. Its support is $[0, 11]$, and the displayed coefficients show strict positivity throughout the interior. Its integral is $P(1) = 1$. This proves order nine is a probability density. Every later order is its convolution with additional uniforms, and the sign under any common tilt is unchanged by (7). $\square$

j	$b_{j,0}$	$b_{j,1}$	$b_{j,2}$	$b_{j,3}$	$b_{j,4}$	$b_{j,5}$	$b_{j,6}$	$b_{j,7}$	$b_{j,8}$
0	0	0	0	0	0	0	0	0	1
1	1	2	4	8	16	32	64	128	246
2	246	364	536	784	1136	1624	2276	3094	4047
3	4047	5000	6088	7280	8512	9680	10648	11300	11573
4	11573	11846	11740	11192	10192	8816	7240	5672	4293
5	4293	2914	1724	904	608	904	1724	2914	4293

Table 1: Degree-eight Bernstein coefficients of $8!f_9(j+t)$ on $0 \leq t \leq 1$. Symmetry about $11/2$ supplies the other five cells.

At the exact threshold the source has ten h -cap coordinates and one $6h$ -cap coordinate, while the target has two coordinates. The polynomial still has a negative coefficient. This shows why coefficientwise positivity was necessary in the equal-cardinality theorem of [4] but is not necessary after uniform smoothing.

For comparison, the first nonnegative half-step coefficient certificate occurs at $N = 11$. The coefficients of $(1+u)^{11}(1-u^2+u^4)$ are

$$(1, 11, 54, 154, 276, 308, 187, 33, 33, 187, 308, 276, 154, 54, 11, 1).$$

This proves a sufficient bound through (12); the spline certificate improves it to the optimal order nine.

6 Attribution and interpretation

The eventual coefficient-positivity mechanism is Pólya’s theorem. Powers and Reznick [1, 2] give primary accounts and quantitative refinements. Handelmann [3], introduction pages 2–3, discusses eventual positivity of finite signed lattice measures under convolution powers and traces the discrete theory to earlier work. These are substantial antecedents; the present note makes no broad priority claim for eventual positivity.

The use here is the half-unit decomposition (12), the resulting exact stabilization criterion for commensurate observation masks, and the explicit minimal-order spline certificate. The fixed-order result extends the preceding mask report [4] to unequal cardinalities. All conclusions concern one kernel valid throughout the total-sum weight or threshold family. They do not assert that every individual two-hypothesis comparison has the same obstruction. The proofs are ordinary mathematical arguments, not formalized or externally refereed results.

References

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- [3] D. Handelmann, *Matrices of positive polynomials*, Electronic Journal of Linear Algebra **19** (2009), 2–89. doi:10.13001/1081-3810.1348. Publisher full text.
- [4] *Universal Comparison of Fabius Observation Masks*, research companion prepared for Vladimir Reshetnikov with OpenAI, 1 October 2026. Source `universal_fabius_mask_criterion.tex`, Theorems 1.1 and 3.1. The relevant ProveIt source question is pinned in that report at commit 7421a4ca60fd.